

Describing Meaning – Week 12

Handout: Modality

w_0 is the real world.

some $\rightsquigarrow \exists x$ every $\rightsquigarrow \forall x$

possibility $\rightsquigarrow \exists w$ necessity $\rightsquigarrow \forall w$

- **epistemic**: given what is known, inferred, or believed
- **deontic**: given rules, duties, permissions, or morals
- **circumstantial**: given the relevant circumstances
- **bouletic**: given desires or wishes
- **teleological**: given a goal

1. Modal flavour

1.1 Modals in Context

For each sentence below, identify the most natural modal flavour of the bolded expression.

- (a) Harry **needs to** return the book by noon.
- (b) The last of the sunlight faded to our left, so we **must** be facing north.
- (c) The prophecy **could** be true.
- (d) According to the teacher's decision, Mario **cannot** go on the class trip.

1.2 Lexical items across contexts

For each sentence below:

- say whether the modal expression allows an **epistemic** reading, a **deontic** reading, **both**, or **neither**,

- if it allows both, give one context for an epistemic reading and one for a deontic reading,
- if it allows neither, say which flavour is more appropriate.

- (a) *Theo may be in the archive.*
- (b) *Leila should leave after lunch.*
- (c) *Nadia must submit the form by noon.*
- (d) *Nadia might be on the 8:15 train already.*
- (e) *With those boots, you can cross the stream.*
- (f) *He has to enter both codes.*
- (g) *Perhaps the parcel is behind the desk.*
- (h) *Ideally, we will finish before sunset.*

2. Modal force

might p is true iff p is true in **some** worlds consistent with my beliefs.

must p is true iff p is true in **all** worlds consistent with my beliefs.

So *might* patterns with $\exists w$, while *must* patterns with $\forall w$.

For each expression below, say whether it patterns more naturally with **existential** or **universal** quantification over worlds. If you think the classification is less straightforward, say why.

- (a) *possibly*
- (b) *necessarily*
- (c) *may*
- (d) *can*
- (e) *have to*
- (f) *should*
- (g) *ought to*
- (h) *think*
- (i) *know*

3. De re and de dicto

A **de re** reading is one on which some individual or time is fixed outside the intensional environment.

A **de dicto** reading is one on which the descriptive or temporal material is interpreted inside the intensional environment.

- (a) *Clara might marry a violinist.* (modal auxiliary)

de re: There is a particular violinist such that Clara might marry that person.

de dicto: It might be that Clara marries some violinist or other.

- (b) *Jonah believes that every reviewer praised his paper.* (attitude verb)

de re: For each particular reviewer, Jonah believes that that reviewer praised his paper.

de dicto: Jonah believes that every reviewer, whoever counts as a reviewer in his belief worlds, praised his paper.

- (c) *In 1975, the dean was a child.* (time)

de re: A particular contextually salient dean was a child in 1975.

de dicto: Whoever counted as dean in 1975 was a child in 1975.

For each sentence below, give a **de re** paraphrase and a **de dicto** paraphrase if both are available. If only one reading seems available or natural, say so.

- (a) *Possibly, Nina will interview a dean.* (modal adverb)

- (b) *Omar is looking for a translator.* (intensional transitive)

- (c) *Sara hopes to marry a dentist.* (attitude verb)

- (d) *Mira thinks that every violinist in the orchestra is backstage.* (attitude verb)

- (e) *A week ago, Mary thought the exam was tomorrow.* (time)

4. Scope

4.1 Nominal quantifiers

As a reminder, for each sentence below:

- give the **surface-scope** reading,
- say whether an **inverse-scope** reading is available,
- paraphrase both readings if both are available.

- (a) *A reviewer read every abstract.*

- (b) *Every candidate consulted a supervisor.*

- (c) *Some photographer followed every cyclist.*

4.2 Modal scope and formal representation

Let c stand for Clara and let $Acc(w_0, w')$ mean that w' is accessible from w_0 .

Consider again:

Clara might marry a violinist.

A **de re** representation is:

$$\exists x \left[\text{violinist}(x, w_0) \wedge \exists w' (Acc(w_0, w') \wedge \text{marry}(c, x, w')) \right]$$

- (a) Give a **de dicto** representation for the same sentence.
- (b) What changes between the de re and de dicto formulas?
- (c) In which world is **violinist** evaluated on each reading?

Now let j stand for Jonah, let p_j stand for Jonah's paper, and let $Bel_j(w_0, w')$ mean that w' is compatible with Jonah's beliefs in w_0 .

Consider:

Jonah believes that a reviewer praised p_j .

A **de re** representation is:

$$\exists x \left[\text{reviewer}(x, w_0) \wedge \forall w' (Bel_j(w_0, w') \rightarrow \text{praised}(x, p_j, w')) \right]$$

- (d) Give a **de dicto** representation for this sentence.
- (e) What changes between the de re and de dicto formulas?
- (f) Does the individual quantified over have to be fixed in the actual world on both readings?

Negation and modal auxiliaries

For each sentence below:

- say whether the natural reading has **negation above the modal** or **negation below the modal**,
- say whether the other scope configuration is available,
- if not, give a clearer English paraphrase for the unavailable reading.

- (a) *Priya must not share the password.*
- (b) *Guests can not use the side entrance.*
- (c) *Omar cannot park in the courtyard.*